TIMER, now in Python

What it does, what the buttons mean, and why you can stop opening MATLAB

Short version: the MATLAB scan script, the Svirgo timing script and the fitting notebook are now one window. Same numbers as before (really the same, see the last section), a few bugs less, and the plots finally have some colour.

Everything below was done on our SrO Scan_002 and on the YAG timing scans, so what you see is real data, not a demo.

Contents

1 Starting it **2**

2 A scan, start to end **2**

2.1 What we typed in . . . 2

2.2 While it runs . . . 2

2.3 At the end . . . 3

2.4 The other two tabs . . . 4

2.5 The buttons . . . 5

3 Backgrounds: OnlyProbe, NoProbe, or neither **5**

4 Error bars, done properly **6**

5 Fitting the trace **7**

5.1 Thermal decay, and the bouncer . . . 7

5.2 FFT of what is left . . . 8

5.3 One or two frequencies? . . . 8

6 Timing (Svirgo), with a peak finder **9**

6.1 Finding the spot . . . 9

6.2 A vs B . . . 10

7 Can we trust it? **10**

1 Starting it

1. Copy the `TIMER_python` folder on the beamline PC. Keep the `.py` files together, they import each other.
2. Open `timer_gui.py` in Spyder and press F5. That's it, Anaconda already has everything.
3. If Spyder complains about the event loop: *Run* → *Configuration per file* → *Execute in a dedicated console*. Spyder and Tk don't love each other.

You can also double-click `start_timer_gui.bat`, or run `python timer_gui.py` from a prompt.

2 A scan, start to end

2.1 What we typed in

This is the SrO run used for every picture of this section:

Scan	SrO, Scan_002, 84 files, FEL delay as x axis
T0	191
ROI (col, row, w, h)	406, 431, 50, 50
Ellipse axes / centre in ROI	20, 28 / 30, 26 (fixed)
Proximity background	30 pixels on each side
Spectrometer window	pixels 600–800
Normalisation	$I0^3$
Background	mean of the first 5 signal images
CCD counts per photon	36 (more on this in Section 4)

One warning from experience: if the spectrometer window is where the spectrometer sees nothing, every shot is thrown away and the program now tells you so. With the default 110–400 on this spectrometer you get exactly that.

2.2 While it runs

Each file is shown for a moment before the next one (*Show each file for at least*, default 0.3 s), so you actually see the scan growing instead of only the final result. Set it to 0 if you are in a hurry.



Top row: shot-to-shot I0 (rejected shots in red), the spectrum, the full CCD image with the ROI and the proximity frames. Middle: the zoom with the ellipse, then the TG signal against file number (with the delay on the top axis) and against time. The shaded band is the error bar, explained in Section 4.

2.3 At the end



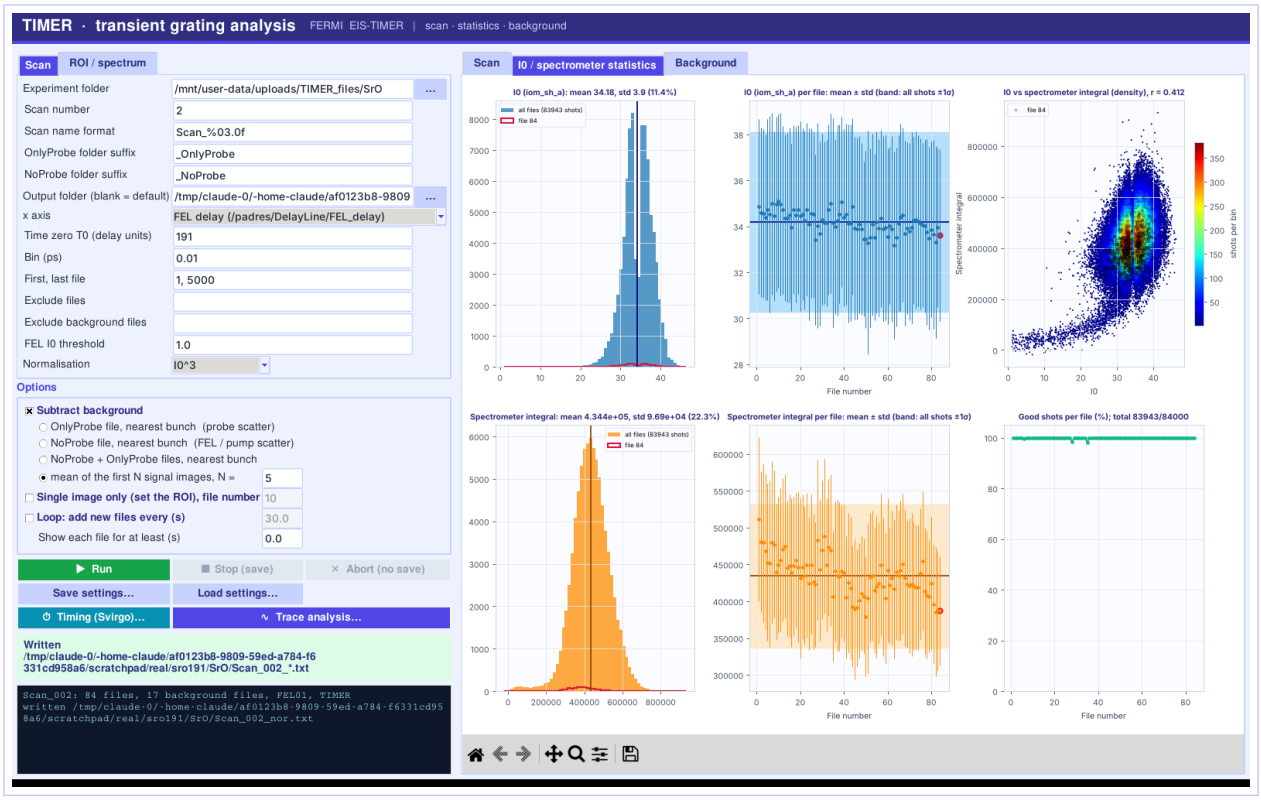
The bottom middle plot is the binned trace that goes into `Scan_002_nor.txt`. With these settings

the oscillation peaks at about 10 counts per shot.

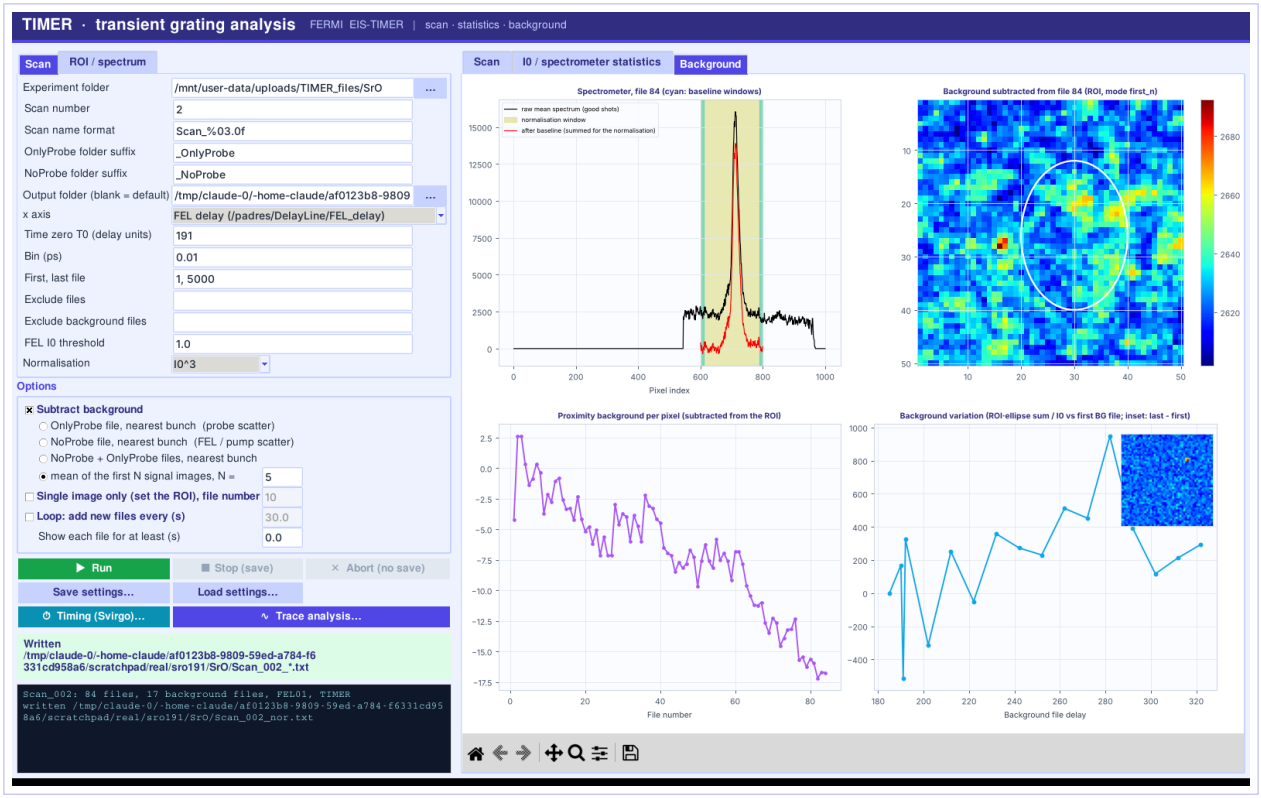
About the images: they are linear, but smoothed by 1.5 pixels and coloured from the median (blue) to the 99.95th percentile (red). Only for the eyes, the analysis uses the raw counts. With log scale the TG spot just drowned in the probe disc.

2.4 The other two tabs

I0 / spectrometer statistics piles up every good shot of every file, so you see at once if the FEL is drifting or the spectrometer is unhappy. The I0 vs spectrometer plot is a density map, because 80 000 grey dots on top of each other tell you nothing.



Background has the spectrum with the integration windows, the background image that was subtracted, the proximity background file by file, and how stable the background files were during the scan.



2.5 The buttons

Run	Go. With *Loop* ticked it keeps picking up new files while the scan is still running.
Stop (save)	Finishes the current file, then bins and saves what it has.
Abort (no save)	Stops and writes nothing. For when you notice the wrong scan number.

Single image only analyses one file and shows the full image and the zoom, so you can place the ROI before the real scan. *Save / Load settings* keeps one settings file per sample, and the last settings come back by themselves at the next start.

3 Backgrounds: OnlyProbe, NoProbe, or neither

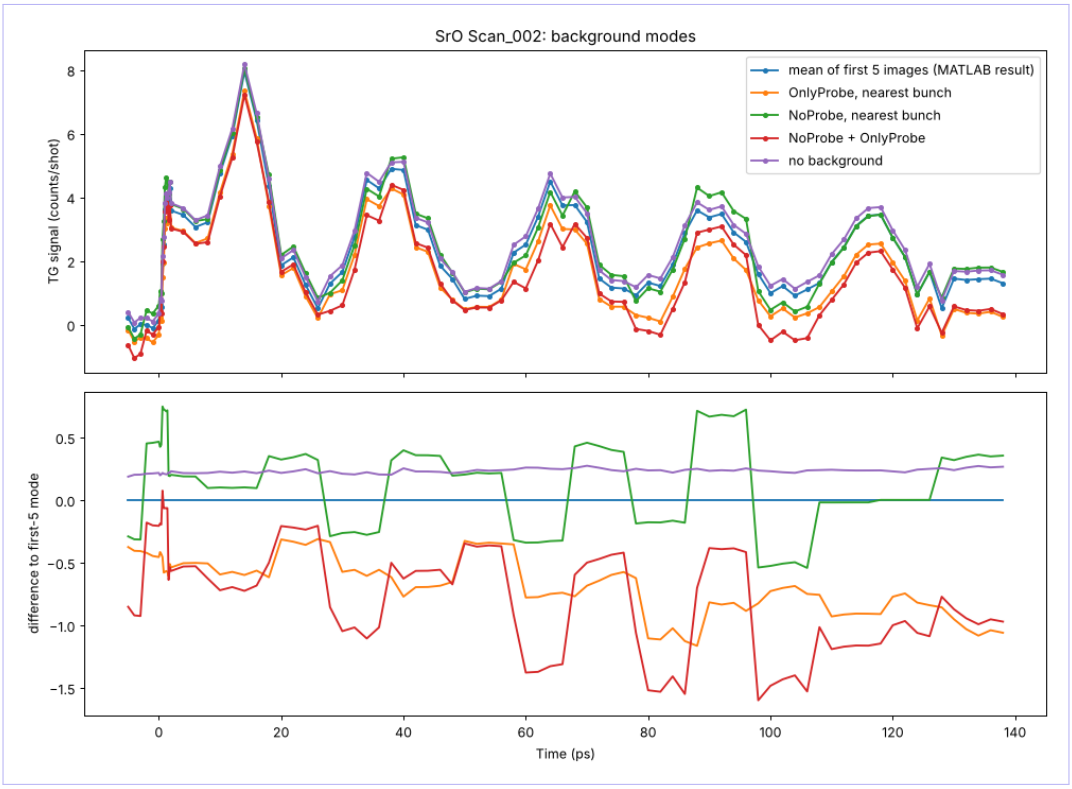
Next to every `Scan_002` there are `Scan_002_OnlyProbe` (probe on, FEL off: probe scatter) and `Scan_002_NoProbe` (FEL and pumps on, probe off: FEL and pump scatter). You can subtract:

- the OnlyProbe image closest in bunch number to each file (MATLAB `BG_Sig = 0` with `_OnlyProbe`),
- the NoProbe image closest in bunch number (same, with `_NoProbe`),
- both,
- or the mean of the first N signal images (MATLAB `BG_Sig = N`).

A small MATLAB confession

`GetBGbyBunchNumber.m` reads the bunch number from the wrong piece of the file name. With the old names it worked; with `Scan_002_<bunch>.h5` it reads "002" for every file, so MATLAB always used the *first* background file. The Python version uses the real nearest one.

Which one is best? We tried all of them on SrO (with the earlier test settings):



The NoProbe images carry about 25 extra counts per pixel of fairly flat FEL light. The proximity background already removes a flat offset, so subtracting a NoProbe image mostly adds its noise: signal-to-noise drops from 65 (first 5 images) to 23. OnlyProbe sits in between (52). For SrO we keep the first N images. Your sample may disagree, and it is one click to check.

4 Error bars, done properly

Each h5 file is one CCD image integrated over about 1000 shots, so there is no shot-by-shot spread to take a standard deviation of. The error is built from what we know about the image:

- **pixel noise**: the spread of the pixels in the four proximity frames, after background subtraction;
- **the proximity background** we subtract also has its own error, and it counts for every pixel of the ellipse;
- **shot noise of the signal itself**: TG counts divided by the counts that one photon gives;
- **the normalisation**: the standard error of the I0 (and spectrometer) average over the good shots.

$$\sigma_{\mathrm{TG}}^2 = N_{\mathrm{ell}}\,\sigma_{\mathrm{px}}^2 + N_{\mathrm{ell}}^2\,\sigma_{\mathrm{prox}}^2 + g\cdot \mathrm{TG}, \qquad \frac{\sigma_{\mathrm{nor}}}{\mathrm{TG}_{\mathrm{nor}}} = \sqrt{\left(\frac{\sigma_{\mathrm{TG}}}{\mathrm{TG}}\right)^2 + \left(\frac{\sigma_N}{N}\right)^2}$$

Files in the same time bin are combined as $\sqrt{\sum \sigma_i^2}/n$.

Counts per photon (g). Nobody knew the camera model, so we measured it from the data: compare each point with a smooth curve through its neighbours and ask how much signal-dependent noise is needed to explain the scatter. Without shot noise the error bars are too small by a factor $\sqrt{2.8}$; with $g = 36$ (range 27–47) they match. The button *Estimate counts per photon from the last scan* redoes this on any long enough scan. It is an "effective" number: anything that grows with the signal ends up in it, which is what you want for honest error bars.

Before t_0, where there is no signal, the predicted error is 0.15 counts per shot and the measured scatter is 0.14. Good enough to believe it. The errors go to `Scan_002_err.txt`; the MATLAB files are untouched.

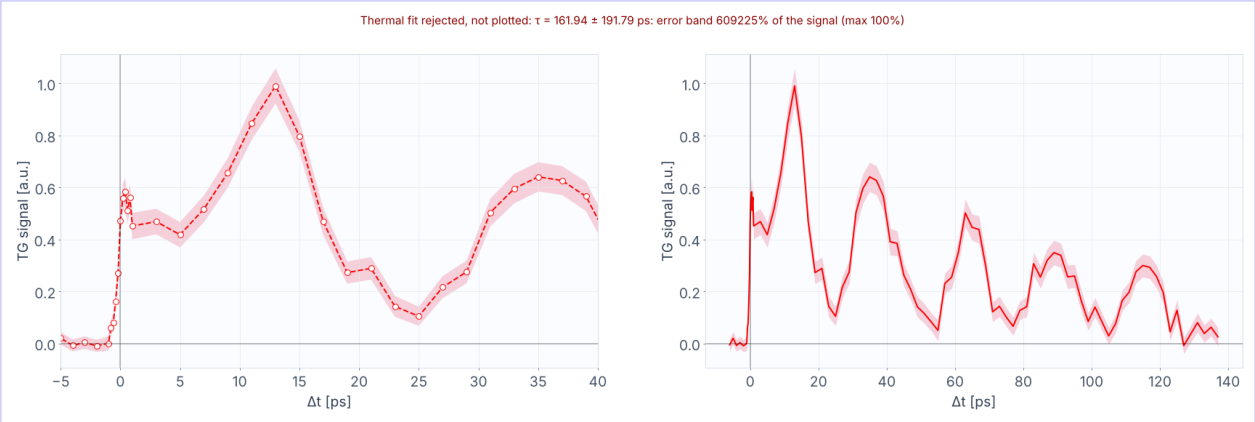
5 Fitting the trace

Trace analysis… opens the last `_nor.txt` and fits it with exactly the notebook models.



5.1 Thermal decay, and the bouncer

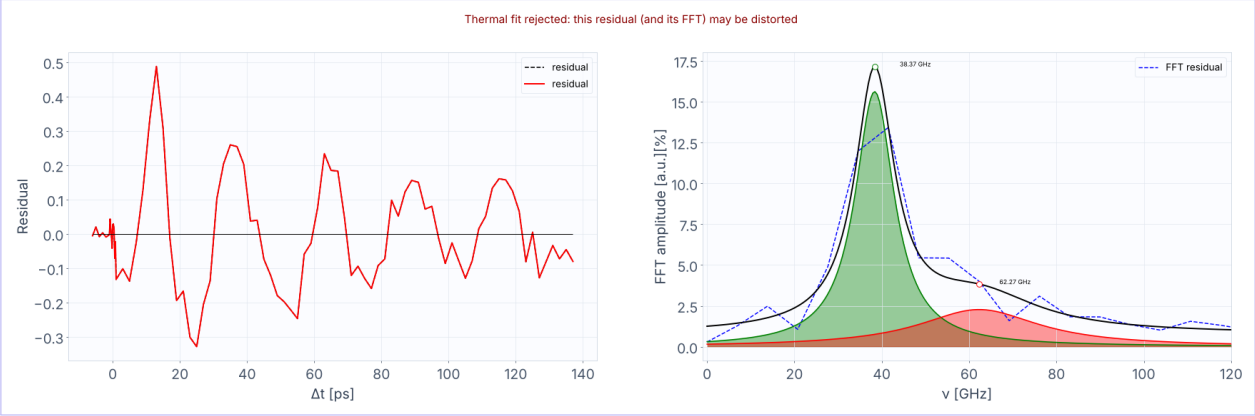
First the thermal background $\left|\frac{1}{2}\left(1+\operatorname{erf}\frac{t-t_0}{\sigma}\right)\left(a\,e^{-(t-t_0)/\tau}+y_0\right)\right|^2$. On this scan the fit gives $\tau = 162 \pm 192$ ps: an error bigger than the value. The filter rejects it and does not draw it, and says why on top of the plot:



A fit is rejected when a parameter sits on its bound, when the τ error band is wider than the signal, or when $R^2 < 0.3$. Tick *Plot the thermal fit even when rejected* if you want to look at it anyway. The pink band on the data is the error bar from Section 4.

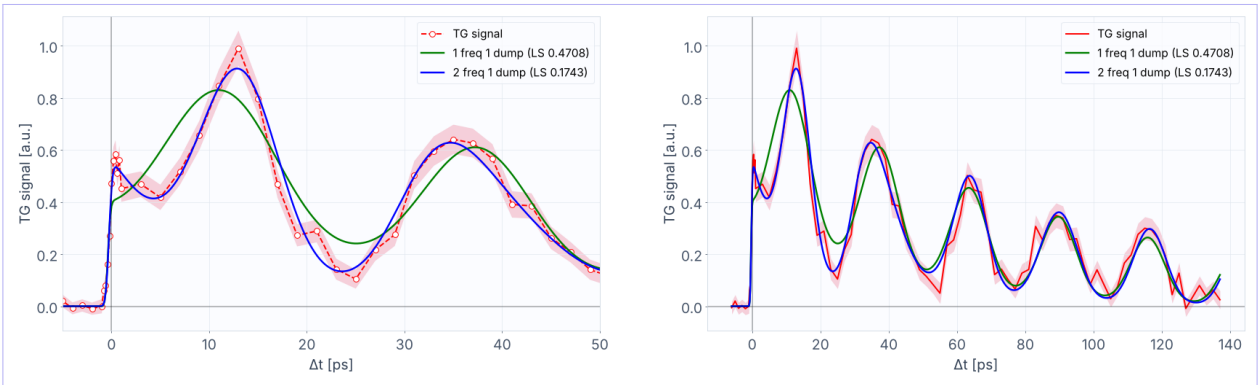
5.2 FFT of what is left

The residual is Fourier transformed and fitted with Lorentzians, which give the starting frequencies. Our time steps are not even (0.2 ps around t_0, 2 ps later), so the FFT is computed on the real time points instead of pretending the step is constant. On a uniform scan it is the same as the notebook.



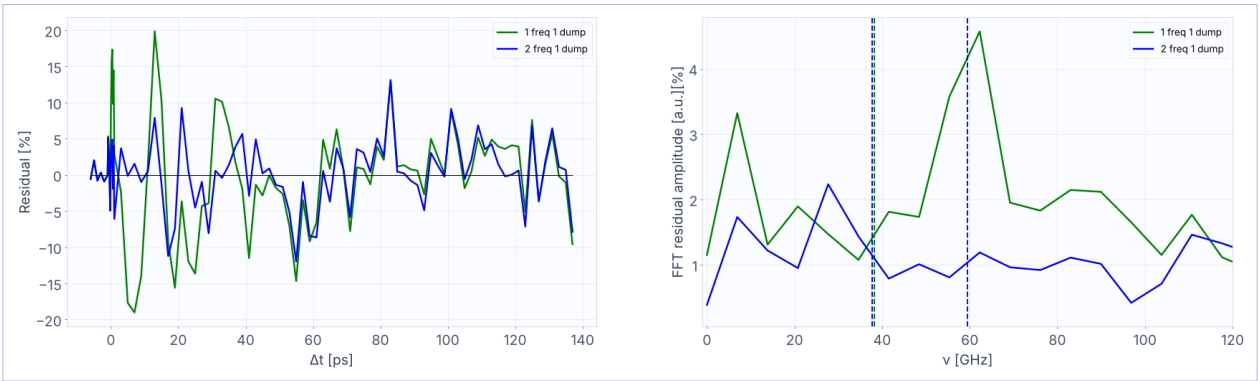
5.3 One or two frequencies?

Tick both and they are overlaid:



Model	ν_1 (GHz)	ν_2 (GHz)	least squares
1 freq 1 dump	38.25 ± 0.33	–	0.47
2 freq 1 dump	37.70 ± 0.19	59.5 ± 1.4	0.17

The second frequency is not decoration: it fixes the shape of the first peak and the residual drops by almost a factor of three.



6 Timing (Svirgo), with a peak finder

First thing at the beamtime: are pump A and pump B on time with each other? Each pump is scanned on the YAG, and the FEL-induced change of the laser transmission is followed in a ROI as a function of the seed delay line.

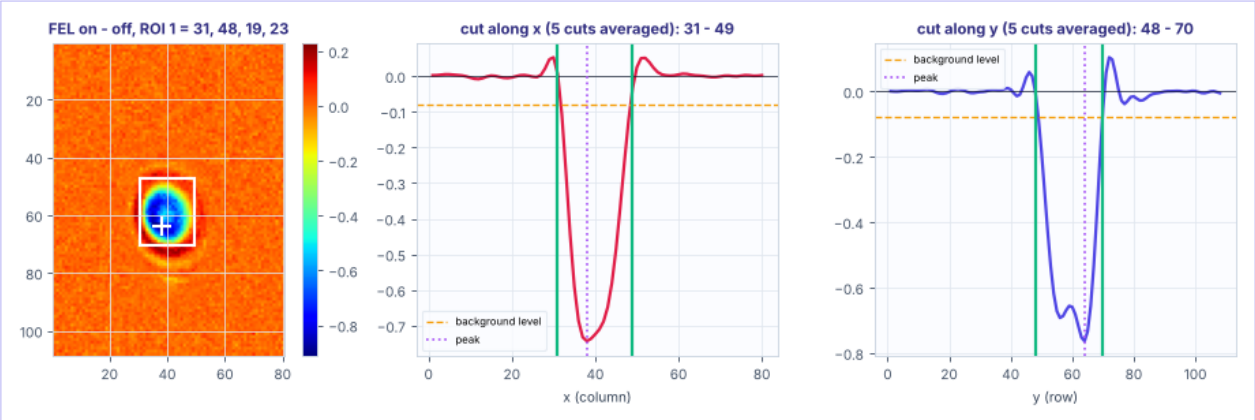
6.1 Finding the spot

The camera image is 108×80 pixels and the FEL spot is about 20×20. The old default ROI (75, 60, 70, 70) is outside the image, which explains a lot of noisy timing traces.

Find the peak: automatic ROI does this:

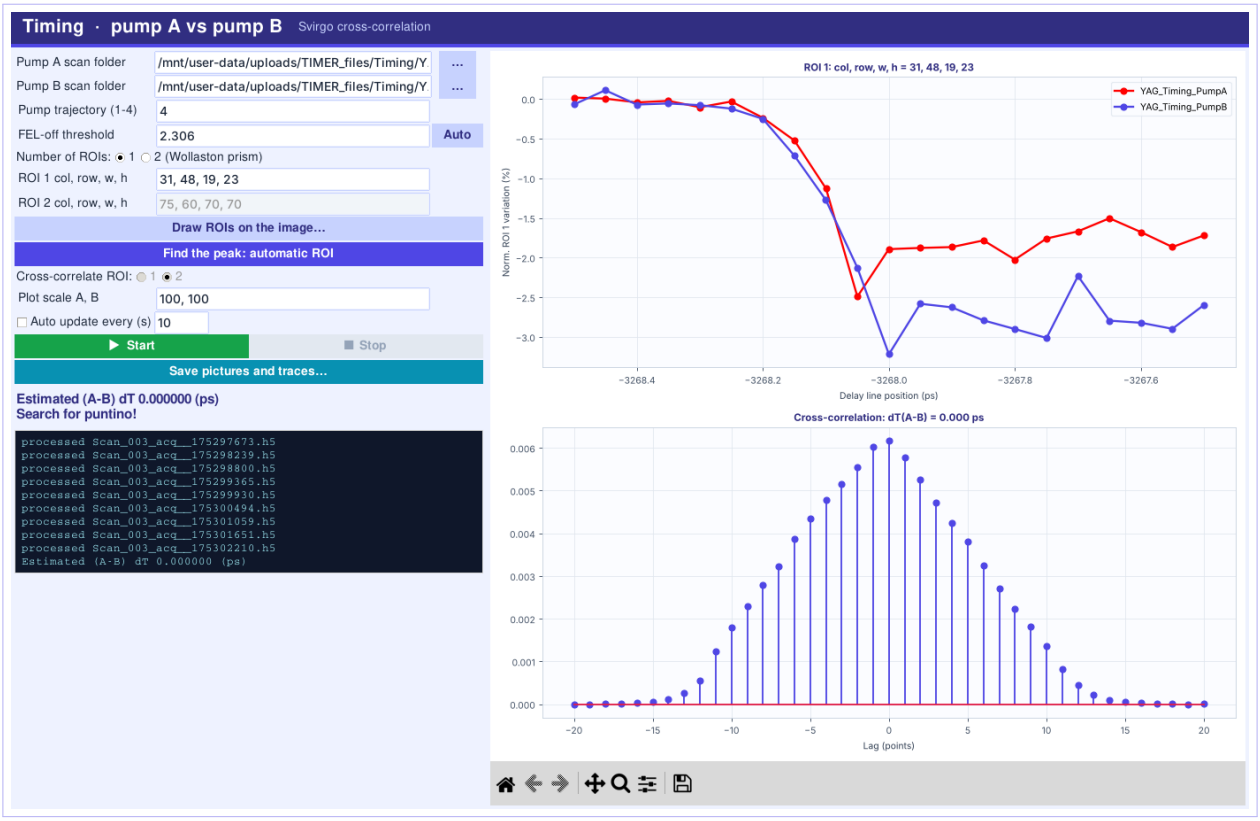
1. builds the FEL-on minus FEL-off image from the files where the spot is strongest;
2. takes the largest change as the peak;
3. walks out from the peak along 5 parallel cuts in x and 5 in y, until the signal drops to the background level (3 times the noise, or 10% of the peak, whichever is larger);
4. the median edge of the cuts, plus one pixel, is the ROI.

You can check it: the window shows the image with the ROI, and the two averaged cuts with the ROI limits as green lines.



If you prefer to do it by hand, *Draw ROIs on the image* now draws on the same on-minus-off image, so you see what you are selecting. With the Wollaston prism (2 ROIs) the finder masks the first spot and looks for the second.

6.2 A vs B



Pump A (Scan_006) and pump B (Scan_003) with their automatic ROIs. Both start dropping at about -3268.15 ps. The cross-correlation says $\Delta T(A-B) = 0$, hence the famous *"Search for puntino!"*. Looking at where each step is half way, they differ by about 0.02 ps, below the 50 fs step of the scan. B drops deeper (-3% against -2%), which says something about the overlap, not about the timing.

Save pictures and traces writes `timing_traces.png`, `timing_roi.png` and the traces as text, ready for the logbook.

7 Can we trust it?

The scan engine was run with the same settings as MATLAB on STO Scan_008 and SrO Scan_002. Every output file (`raw`, `nor`, `pos`, `bg`, `MNF`) is identical, maximum difference exactly zero. The only place where the numbers are meant to differ is the nearest-bunch background, where the MATLAB bug is fixed.

If something looks wrong

- *No good shots*: check the spectrometer window first, then the I0 threshold.
- *The thermal fit vanished*: it was rejected, the reason is printed above the plot.
- *Timing trace is pure noise*: the ROI is probably not on the spot. Press *Find the peak*.
- *Spyder freezes*: dedicated console (Section 1).